

SIEGET[®]25

Low Noise Amplifier with BFP 420 Transistor at 2.4 GHz

The SIEMENS Grounded Emitter Transistor Line is a completely new generation of silicon bipolar junction RF-transistors. This application note describes a low-noise amplifier with the following characteristics:

Gain: **13.6 dB**
Noise Figure: **1.4 dB** @ Frequency=2.4 GHz, $V_{CE}=2.5V$, $I_C=5mA$
 Unconditionally stable at all frequencies

In order to match the input and output ports, a well defined emitter inductance is used as negative feedback. This acts to stabilise the transistor, decreasing the usable gain. Without feedback the gain of the amplifier would be **17.5 dB**, but matching and stability would be degraded. As the feedback is lossless, the amplifier noise figure would remain 1.4 dB.

A double-transistor active stabilisation is used for DC biasing, which provides a temperature stable current source.

Figure 1 shows the circuit diagram:

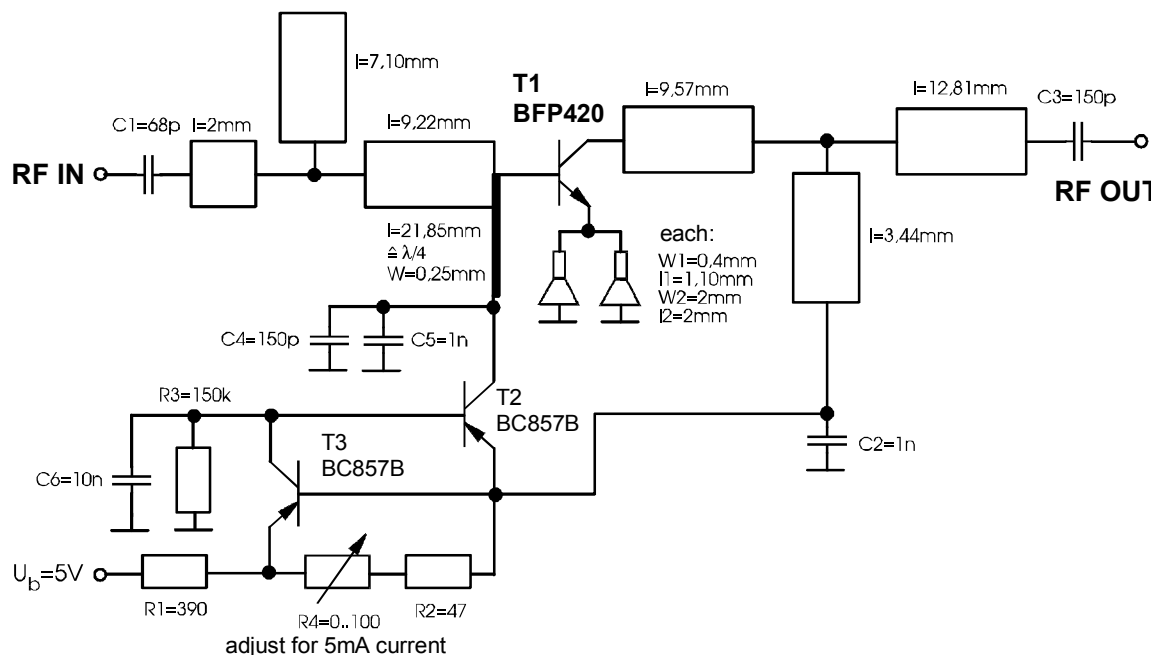


Figure 2 shows the circuit layout. Substrate material is PTFE, $\epsilon_r=2.45$, $H=0.38\text{mm}$, e.g. Di-CLAD 527, 15mil
 Scale: 2:1 (40mm x 40mm real size):

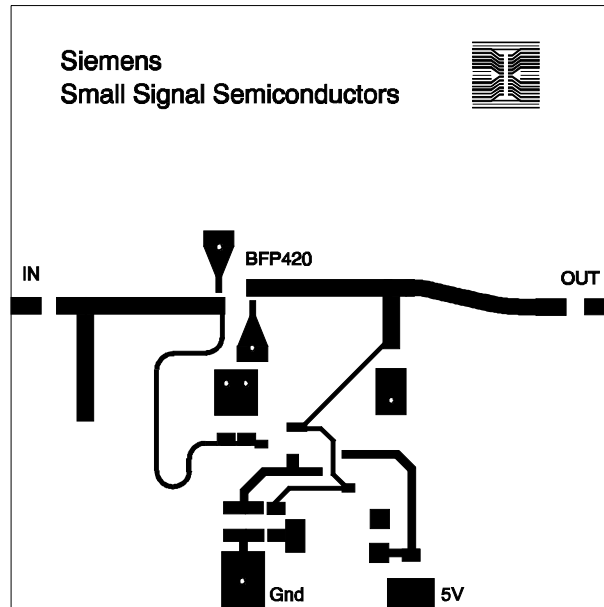


Figure 3 shows the part mounting plan:

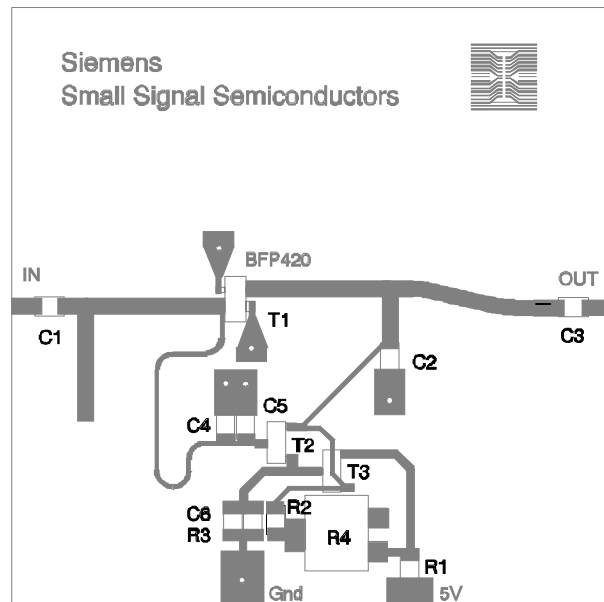
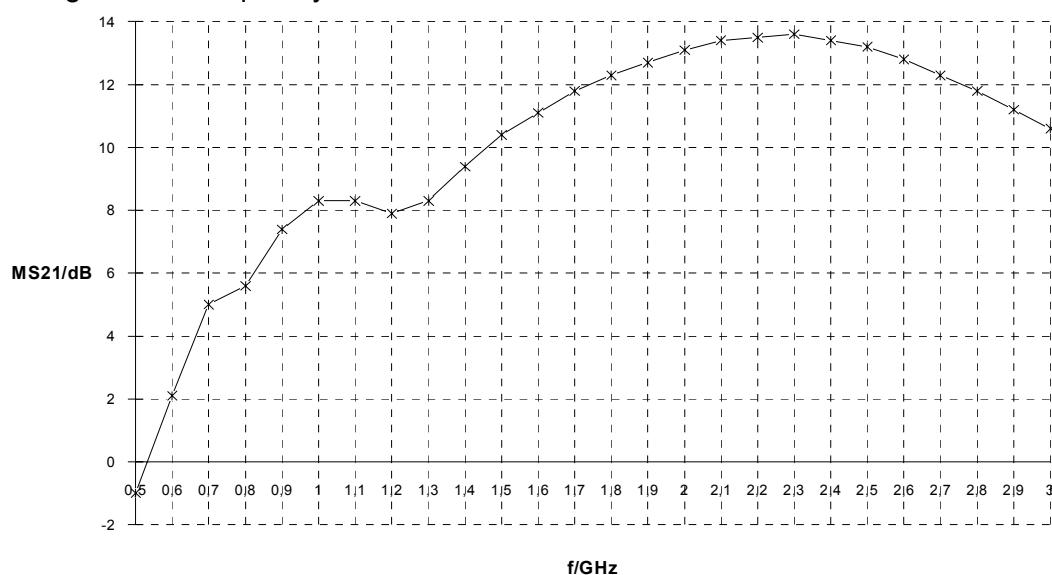


Table 1 lists all used SMD components:

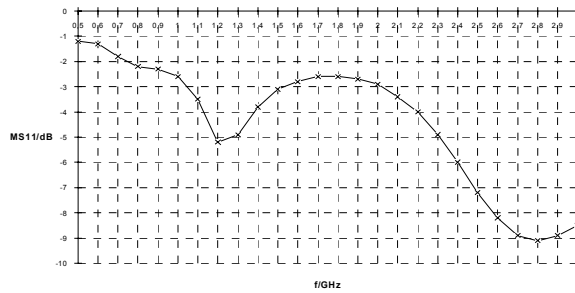
Name	Type, Package, (Supplier)
C1	68pF 0805 (Epcos)
C2	1nF 0805 (Epcos)
C3	150pF 0805 (Epcos)
C4	150pF 0805 (Epcos)
C5	1nF 0805 (Epcos)
C6	10nF 0805 (Epcos)
R1	390 Ω 0805 (Epcos)
R2	47 Ω 0805 (Epcos)
R3	150k Ω 0805 (Epcos)
R4	0...100 Ω Trimmer (Beckman)
T1	BFP 420 SOT343 (Infineon)
T2	BC 857B SOT23 (Infineon)
T3	BC 857B SOT23 (Infineon)

The following diagrams show power-gain, input- and output matching:

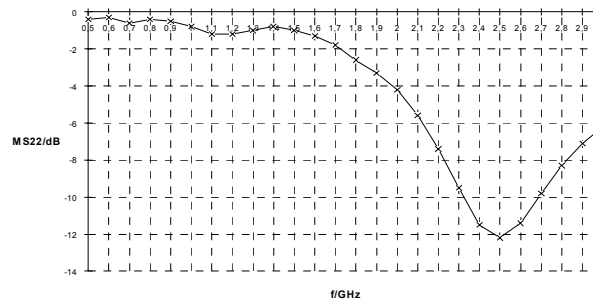
Power gain over frequency:



Input-match:



Output-match:



Measured noise figure:

f/GHz	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
F/dB	1.4	1.4	1.6	1.5	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.5	1.5	1.5	1.4	1.4	1.4	1.4	1.4	1.5	1.6

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